

Why it is difficult: SKU diversity, BOMs and dependent demand

Part 3 · Why is it difficult? — solution report with data, method, results and charts

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1. Why it is difficult: SKU diversity, BOMs and dependent demand

Every end item explodes into many components whose demand is **dependent** — it is *calculated* from the end-item plan, not forecast. Independent demand (end items, spare parts) must be forecast.

18 Distinct purchased components in WM-100 (model BOM)	61 Total component units per machine	2 BOM levels
5,990 12-week machine demand	365,390 12-week component units required	

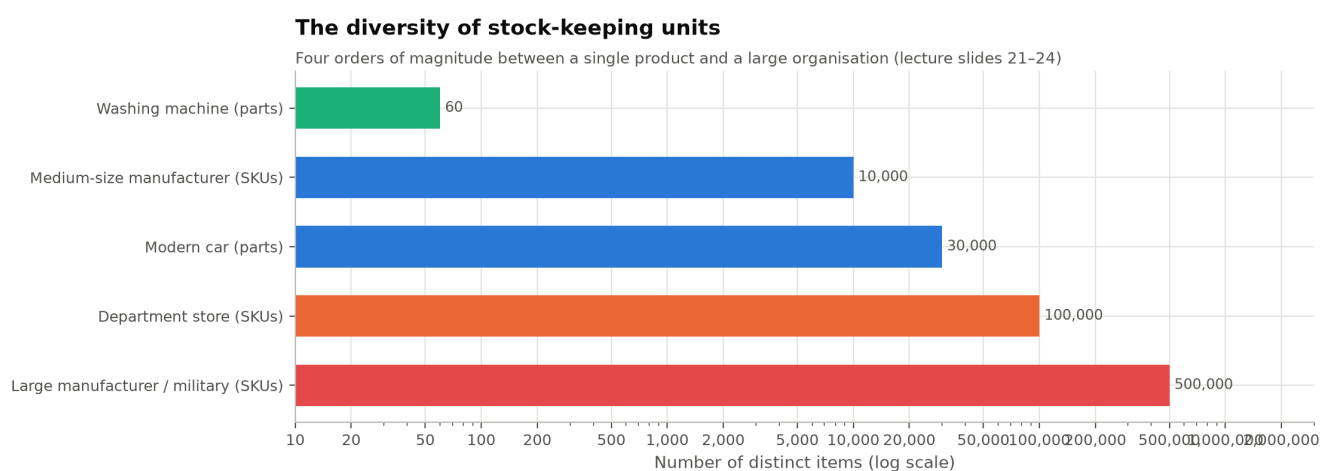


Figure 1.1 — Item counts quoted in the lecture.

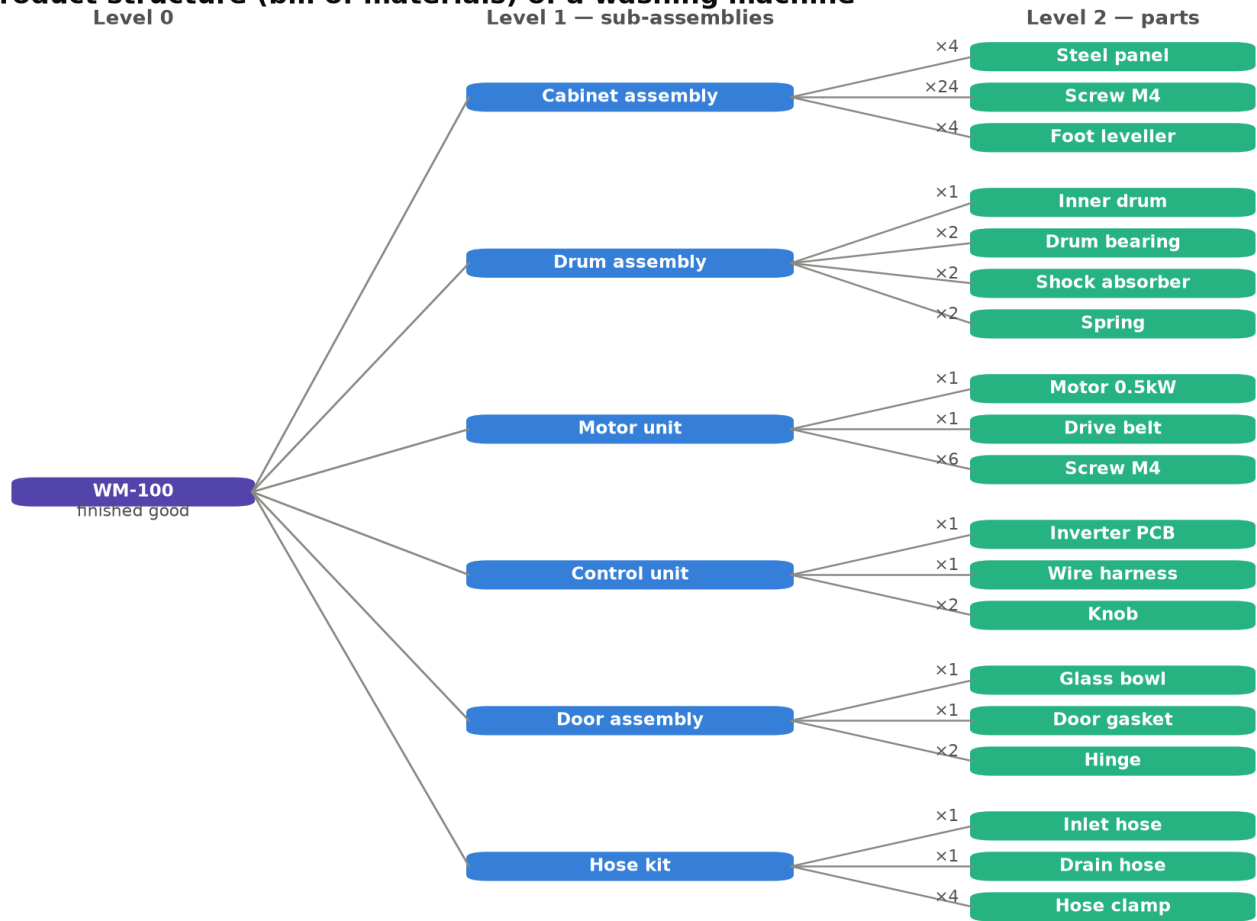
Product structure (bill of materials) of a washing machine

Figure 1.2 — Simplified 3-level BOM. Numbers on the links = quantity per parent.

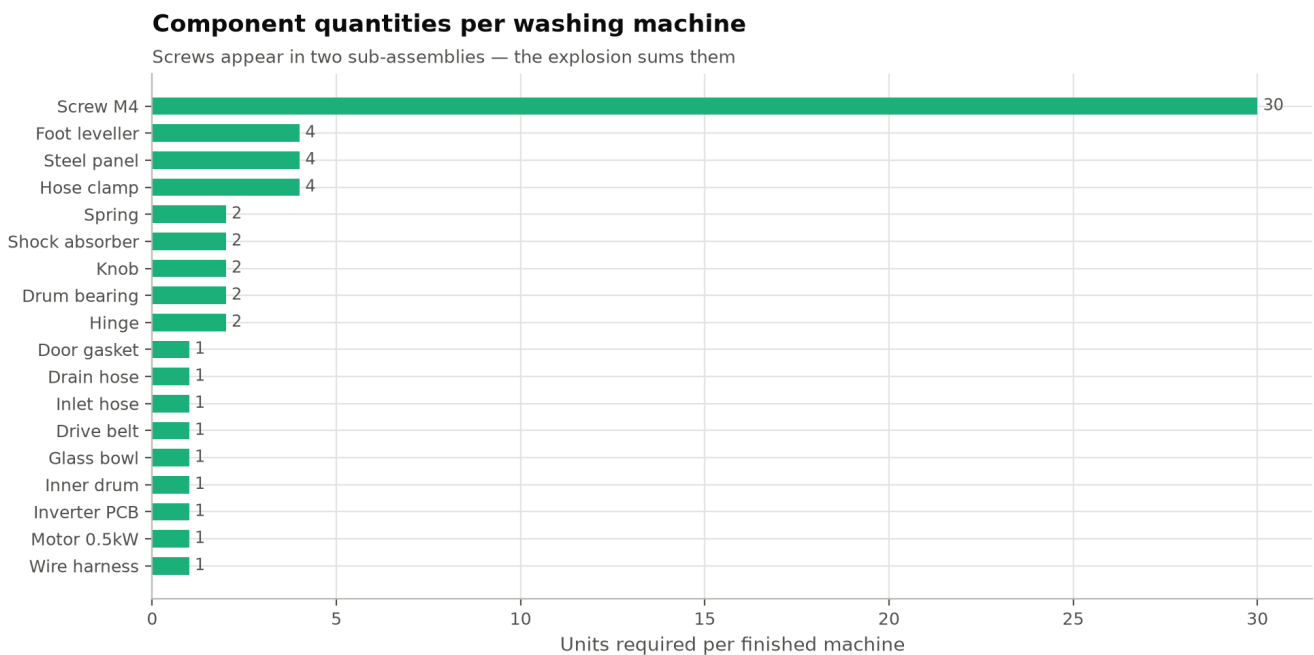


Figure 1.3 — Result of the multi-level BOM explosion.

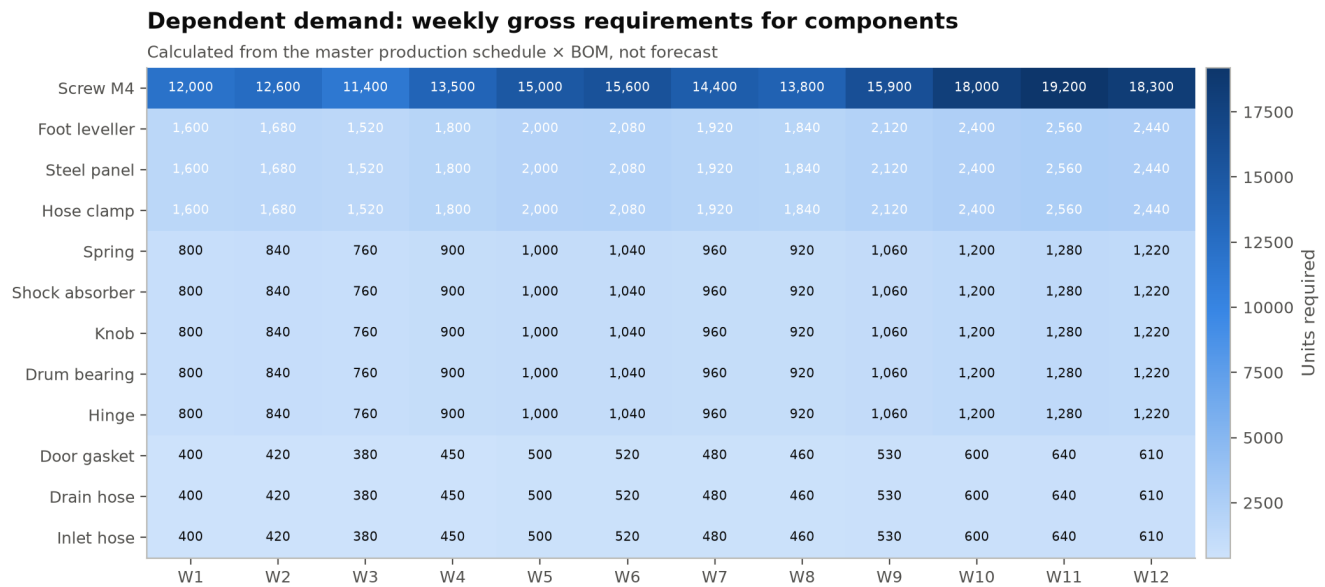


Figure 1.4 — Gross component requirements for the 12 most-used components (lead-time offsetting ignored).

Indented BOM explosion.

level	parent	component	qty per parent	qty per end item
1	WM-100	Cabinet assembly	1	1
2	Cabinet assembly	Steel panel	4	4
2	Cabinet assembly	Screw M4	24	24
2	Cabinet assembly	Foot leveller	4	4
1	WM-100	Drum assembly	1	1
2	Drum assembly	Inner drum	1	1
2	Drum assembly	Drum bearing	2	2
2	Drum assembly	Shock absorber	2	2
2	Drum assembly	Spring	2	2
1	WM-100	Motor unit	1	1
2	Motor unit	Motor 0.5kW	1	1
2	Motor unit	Drive belt	1	1
2	Motor unit	Screw M4	6	6
1	WM-100	Control unit	1	1
2	Control unit	Inverter PCB	1	1
2	Control unit	Wire harness	1	1
2	Control unit	Knob	2	2
1	WM-100	Door assembly	1	1
2	Door assembly	Glass bowl	1	1
2	Door assembly	Door gasket	1	1
2	Door assembly	Hinge	2	2
1	WM-100	Hose kit	1	1
2	Hose kit	Inlet hose	1	1
2	Hose kit	Drain hose	1	1
2	Hose kit	Hose clamp	4	4

Gross requirements by component and week.

component	W1	W2	W3	W4	W5	W6	W7	W8	W9	W10	W11	W12
Screw M4	12,000	12,600	11,400	13,500	15,000	15,600	14,400	13,800	15,900	18,000	19,200	18,300
Foot leveller	1,600	1,680	1,520	1,800	2,000	2,080	1,920	1,840	2,120	2,400	2,560	2,440
Steel panel	1,600	1,680	1,520	1,800	2,000	2,080	1,920	1,840	2,120	2,400	2,560	2,440
Hose clamp	1,600	1,680	1,520	1,800	2,000	2,080	1,920	1,840	2,120	2,400	2,560	2,440
Spring	800	840	760	900	1,000	1,040	960	920	1,060	1,200	1,280	1,220
Shock absorber	800	840	760	900	1,000	1,040	960	920	1,060	1,200	1,280	1,220
Knob	800	840	760	900	1,000	1,040	960	920	1,060	1,200	1,280	1,220
Drum bearing	800	840	760	900	1,000	1,040	960	920	1,060	1,200	1,280	1,220
Hinge	800	840	760	900	1,000	1,040	960	920	1,060	1,200	1,280	1,220
Door gasket	400	420	380	450	500	520	480	460	530	600	640	610
Drain hose	400	420	380	450	500	520	480	460	530	600	640	610
Inlet hose	400	420	380	450	500	520	480	460	530	600	640	610
Drive belt	400	420	380	450	500	520	480	460	530	600	640	610
Glass bowl	400	420	380	450	500	520	480	460	530	600	640	610
Inner drum	400	420	380	450	500	520	480	460	530	600	640	610
Inverter PCB	400	420	380	450	500	520	480	460	530	600	640	610
Motor 0.5kW	400	420	380	450	500	520	480	460	530	600	640	610
Wire harness	400	420	380	450	500	520	480	460	530	600	640	610